

NEW PRODUCTS

TECHNOLOGY

COMPONENTS

Memory fanout gearbox

A new memory interface technology aims to boost data movement between AI processors and memory, addressing a growing bandwidth bottleneck in



modern AI systems. Samsung Electronics has introduced

Weaver, a memory fanout gearbox that expands limited processor memory connections into multiple high-speed channels. Using 112G VSR SerDes lanes, the device enables up to 16TB/s bandwidth while working with standard LPD-DR5X memory. Developed by Credo for AI data-centre hardware, the technology improves data flow without changing core memory architectures. By reducing memory congestion, it can accelerate AI inference workloads and support flexible memory configurations for next-generation AI accelerators.

Samsung Electronics

<https://www.samsung.com/in>

28nm automotive MCU

Renesas Electronics has introduced the RH850/U2C, a 32-bit automotive microcontroller built on a 28nm process to support emerging zonal electrical/electronic (E/E) vehicle



architectures. The MCU integrates four CPU cores running

up to 320MHz, dual lockstep cores for safety-critical workloads, and up to 8MB on-chip flash memory. It also supports high-speed connectivity, including 1Gbps TSN Ethernet, CAN-XL, and multiple legacy automotive interfaces. Developed by Renesas in Japan, the device targets chassis,

safety, battery management, lighting, and motor-control applications while meeting ASIL-D functional safety and advanced cybersecurity requirements.

Renesas Electronics Corporation

<https://www.renesas.com/en>

24-channel mixed-signal IC

Microchip Technology has introduced a highly integrated 24-channel mixed-signal IC designed to simplify actua-



tion control systems in aerospace and defence platforms.

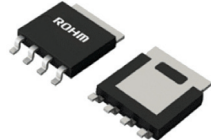
The device combines sensing, monitoring, and motor-control functions into a single chip, helping reduce system size, weight, and component count in mission-critical electronics. The IC also features a redundant architecture for fault-tolerant operation, supporting applications such as more-electric aircraft, drones, guided defence systems, and launch platforms.

Microchip Technology Inc.

www.microchip.com

Chip ferrite beads

ROHM Semiconductor has introduced a compact series of chip ferrite beads designed to improve power supply stability and noise suppression in



automotive electronics. The BLM18SP_SH1 series targets

power circuits used in advanced driver assistance systems (ADAS), radar, cameras, and other in-vehicle electronics. Built using advanced internal electrode fabrication technology, the chips offer improved impedance efficiency with lower resistance, enabling higher current

handling. The components come in a 0603 package—about 50% smaller than conventional designs—allowing greater design flexibility and space savings in modern automotive electronic systems.

ROHM Semiconductor

<https://www.rohm.com>

Surge-protection varistors

TDK has introduced compact ThermoFuse varistors designed to provide high surge protection while reducing component size in power systems.



The new MT40 series integrates a thermal disconnect mechanism and overmolding technology to enhance safety and reliability.

Each device can handle

peak surge currents up to 50kA and supports AC voltages from 150V to 550V. The components target applications such as inverters, industrial power supplies, telecommunications equipment, outdoor lighting, and surge protection devices requiring robust transient protection.

TDK Corporation

<https://www.tdk.com/en/index.html>

Compact voltage-driven photorelays

Toshiba Electronic Devices & Storage Corporation has introduced four compact voltage-driven photorelays



designed for reliable operation in high-temperature

semiconductor testing equipment. The devices—TLP3407SRB, TLP3412SRB, TLP3412SRHB, and TLP3412SRLB—support a maximum operating temperature of 135°C, improving reliability in burn-in systems, probe cards, and semiconductor testers. Housed in a small S-VSON4T package measuring